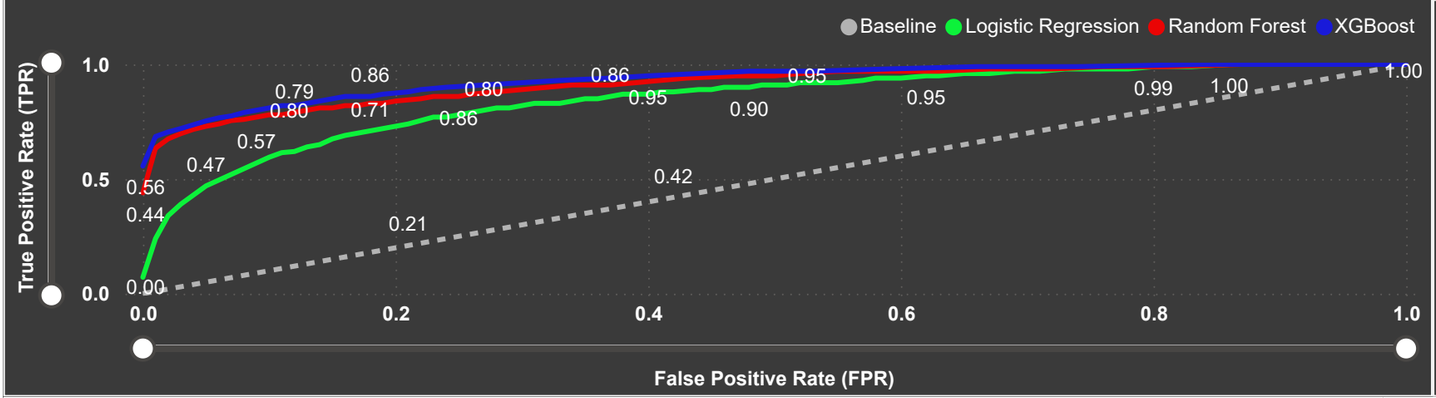
Model Performance Summary

6.335K	1.351K	21.33%	38.47%	26.77%	22.78%
Total Accounts	Total Defaults	Portfolio Default Rate	Pred. PD Logistic Regression	Pred. PD Random Forest	Pred. PD XGBoost

Model	AUC	KS	Precision		Recall		F1 Score	
Logistic Regression	0.85	0.53	48.15% Defaulter	92.09% Non-defaulter	75.28% Defaulter	78.03% Non-defaulter	58.74% Defaulter	84.48% Non-defaulter
Random Forest	0.92	0.68	84.77% Defaulter	92.43% Non-defaulter	70.84% Defaulter	96.55% Non-defaulter	77.18% Defaulter	94.45% Non-defaulter
XGBoost	0.93	0.71	90.22% Defaulter	92.56% Non-defaulter	70.98% Defaulter	97.91% Non-defaulter	79.45% Defaulter	95.16% Non-defaulter

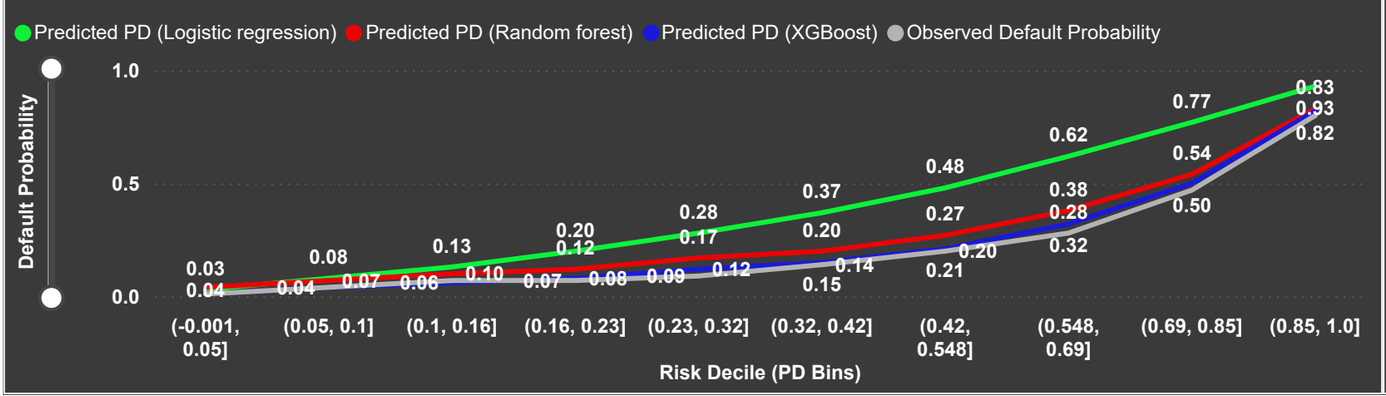
Model Discrimination – ROC Curve



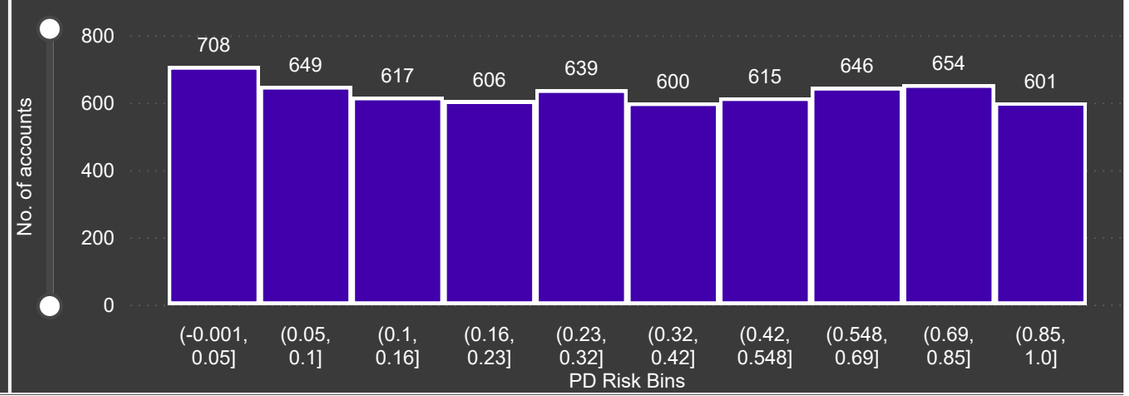
Calibration Table

Risk Bins	Observations	Actual Default	Logistic Regression	Random Forest	XGBoost
(-0.001, 0.05]	708	4	0.03	0.04	0.01
(0.05, 0.1]	649	27	0.08	0.07	0.04
(0.1, 0.16]	617	42	0.13	0.10	0.06
(0.16, 0.23]	606	44	0.20	0.12	0.08
(0.23, 0.32]	639	58	0.28	0.17	0.12
(0.32, 0.42]	600	83	0.37	0.20	0.15
(0.42, 0.548]	615	122	0.48	0.27	0.21
(0.548, 0.69]	646	184	0.62	0.38	0.32
(0.69, 0.85]	654	307	0.77	0.54	0.50
(0.85, 1.0]	601	480	0.93	0.83	0.82
Total	**6335**	**1351**	**3.89**	**2.72**	**2.31**

PD Model Calibration Curve



Distribution of accounts across PD Bins



Confusion Matrix

Logistic Regression

Actual status	Defaulter	Non-defaulter	Total
Non-defaulter	1095	3889	**4984**
Defaulter	1017	334	**1351**
Total	**2112**	**4223**	**6335**

Random Forest

Actual status	Defaulter	Non-defaulter	Total
Non-defaulter	172	4812	**4984**
Defaulter	957	394	**1351**
Total	**1129**	**5206**	**6335**

XGBoost

Actual status	Defaulter	Non-defaulter	Total
Non-defaulter	104	4880	**4984**
Defaulter	959	392	**1351**
Total	**1063**	**5272**	**6335**